# When Search Is Not Enough: Diagnosing Failure Before Constructing Verified Reasoning Capabilities

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CORA on ARC-AGI-2 · development results · ARC Prize Research Summit, 23 October 2026

## The question

**How can a reasoning system know when its existing way of solving problems is inadequate, and acquire a better one without compromising the evidence used to judge it?**

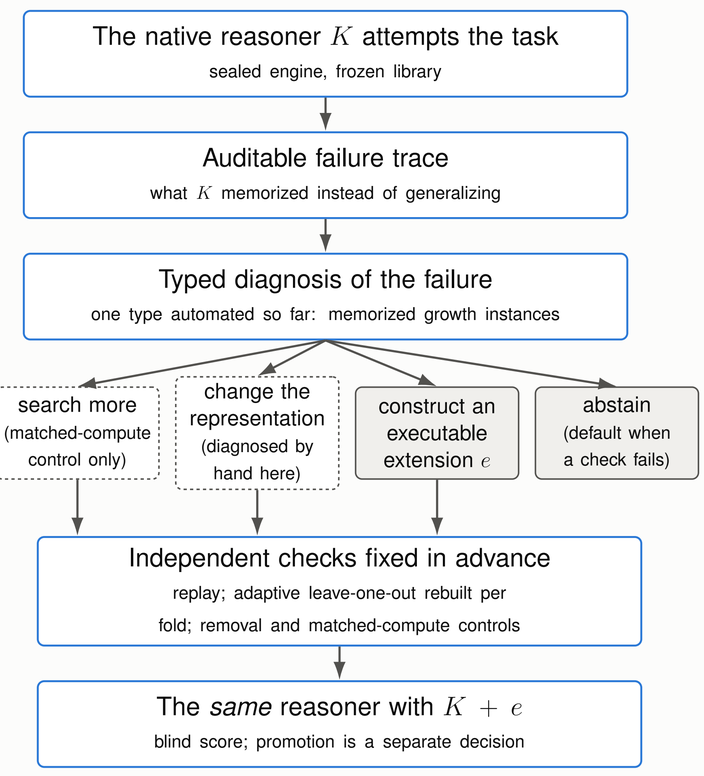
A failure to solve is ambiguous. It can mean:

- **search**: a program exists in the bounded language, but search missed it;
- **representation**: the right object view was screened out before induction;
- **expressivity**: the bounded language holds no solution (needs a complete check);
- **evidence**: the demonstrations support incompatible rules;
- **verification**: a valid candidate is rejected or not reproduced when examples change.

These call for different responses: search longer, change the view, construct an executable extension, or abstain.

*Self-modifying executable knowledge, with non-self-modifying standards of evidence.*

## The CORA loop




Solid boxes are implemented and measured; dashed ones are specified but not automated. Four claims are kept apart and never inferred from each other: **C1** the extension installs and runs; **C2** $K + e$ beats $K$ and matched removal and compute controls; **C3** the real failure trace adds value over demonstrations-only and shuffled-failure construction; **C4** a retained capability helps later unseen tasks.

## Evidence ledger

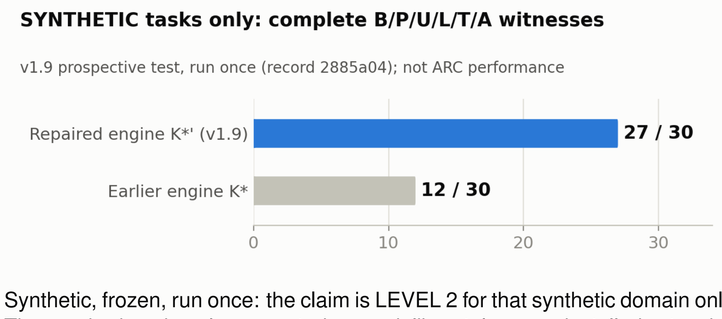
| | |
|---|---|
| synthetic v1.9 prospective (27/30 vs 12/30) | 2885a04 |
| real reach v1 measurement, verified (4/60 installed, 0/14 LOO) | bb11f95 |
| adaptive-LOO diagnosis record | c2b8cd4 |
| segmentation-admission probes | 955542c, 82251a4 |
| representation-eligibility addendum, P2 proposal (not authorized) | a719383 |
| synthetic negative controls; static exposure | 70d3ed9, 3ba75e5 |
| pilot v2 frozen, not run (N = 120) | c1906ee |

Index: `records/CORA_SUMMIT_POSTER_EVIDENCE_20261012.md`

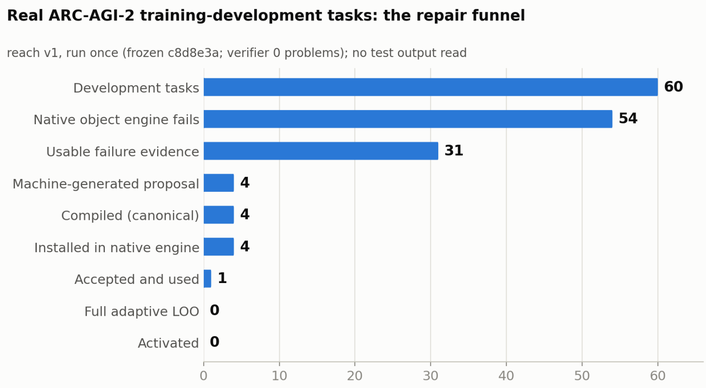
## Take-home

1. A failed solve is not one thing. On real ARC development tasks the blocker we traced was **representation admission** (a search-layer defect in the native engine), not a missing capability.
2. The obvious repair, forcing the right view, was **harmful** on synthetic controls. Negative controls come before any repair is run.
3. Constructed extensions now reach the native engine (0/60 to 4/60 installed; one accepted and used), but **none survives adaptive leave-one-out**; no new ARC solve is claimed.

## What has been measured

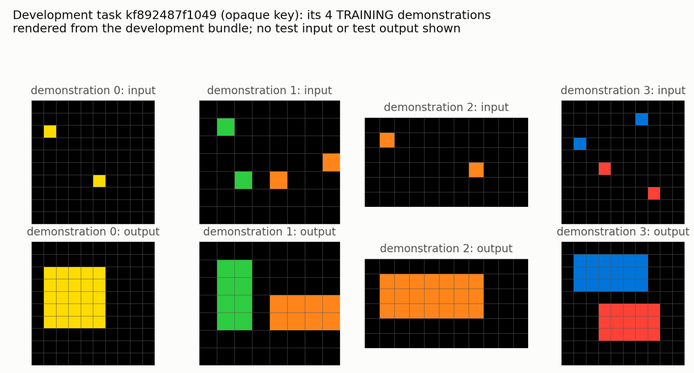



Synthetic, frozen, run once: the claim is LEVEL 2 for that synthetic domain only. The repaired engine also accepted every deliberately wrong installed extension it was given (5/5 and 28/28), so safety must come from task-time verification, not from engine acceptance.

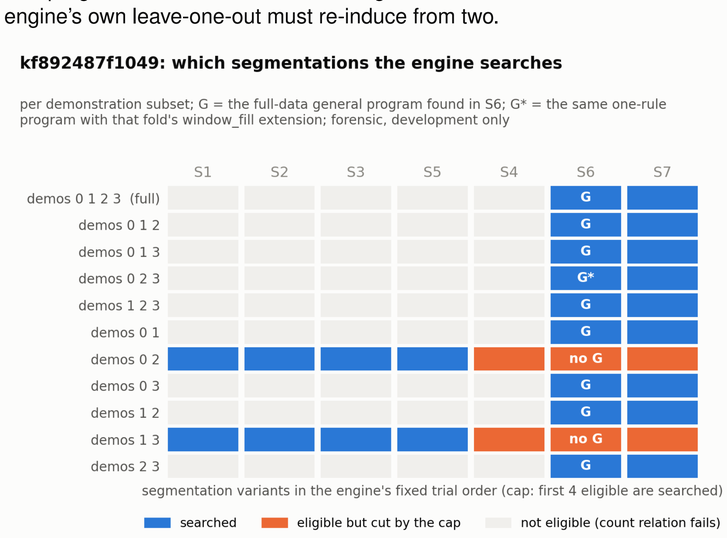



Real ARC-AGI-2 *training-development* tasks, no test output read. Installation rose from 0/60 to 4/60; one repair was accepted and used with all demonstrations; **none survived adaptive leave-one-out (0 of 14 folds); 0 activations**. No new ARC solve is claimed; the certified training score stays 185/1,000. A preserved negative: an earlier failure-association selector did not beat demonstrations alone (0.585 vs 0.591, 288 synthetic groups).

## A forensic counterexample to “just search harder”

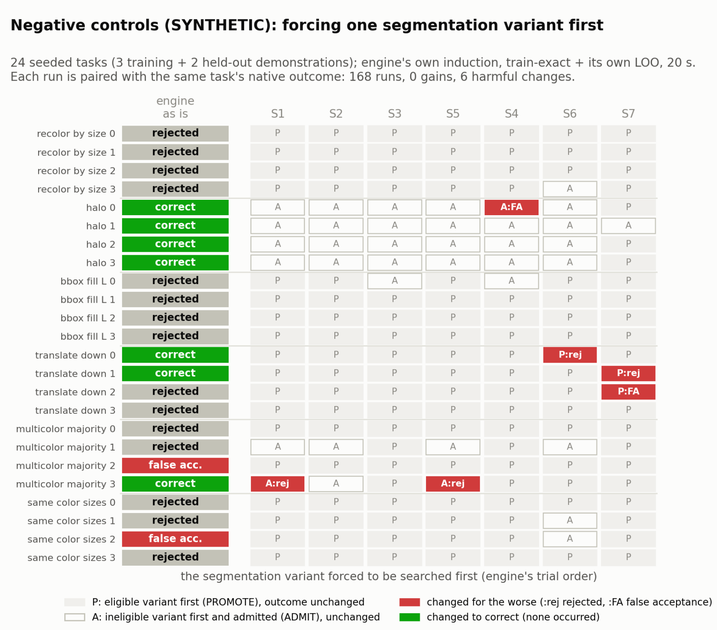



CORA built a `bbox_fill` growth relation from the engine's own memorized failures; with all four demonstrations the unchanged engine accepted the one-rule program $G$ and used it. Rebuilding without one demonstration fails: the engine's own leave-one-out must re-induce from two.



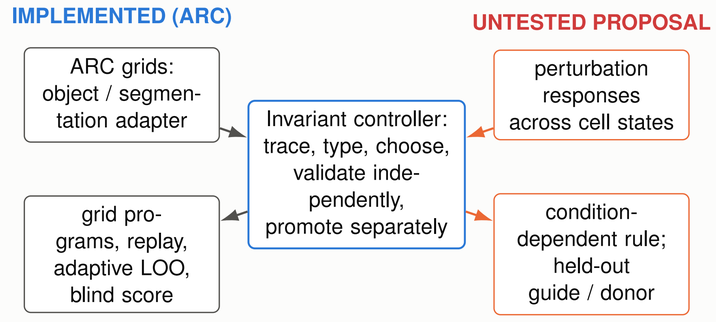

**The mechanism.** The engine searches at most 4 eligible segmentation variants in a fixed order. In the fine variants the demonstrations have object counts $(2 \to 1), (4 \to 2), (2 \to 1), (4 \to 2)$: no common count relation, so they are ineligible, except on the pairs $\{0, 2\}$ and $\{1, 3\}$, where a constant difference makes them eligible and they fill the cap. The correct view (S6) stays eligible and is strictly coarser, but is never searched; $G$ is valid on those pairs yet never generated. The eligibility rule behaves as designed (removing demonstrations never removes an eligible view); the cap defeats the engine's own rule that a coarser view is still tried so that folds converge. Development only; the pairs were examined after the failure was seen.

## Negative controls before any repair




A fix that makes the known task pass proves nothing by itself, so the candidate repairs were first tried on synthetic tasks with known rules and held-out demonstrations. Forcing one segmentation view to be searched first, whether it was eligible or not, produced **no gains and six harmful changes in 168 paired runs**: correct native answers became rejections or false acceptances. We therefore did not run the variant-first repair we had built. Letting the engine reach every cut view would change the search on 73 of 96 synthetic subsets, so that is not narrow either.

## A domain-portable protocol, not a domain-free oracle




The control logic is meant to transfer; observations, program semantics, priors and validators do not. The biological lane has **no data and no results**: it would need leave-one-guide and leave-one-donor tests, calibration, confounder checks and an independent evaluator, and gives predictive validation only (causal biology needs controlled interventions).

## Next falsifiable experiment (not authorized)

1. **P2, leave-one-out anchored completion**: only inside the engine's own leave-one-out, append the validated program's view when a fold finds it eligible but the cap cuts it. No reordering, no admission of an ineligible view, nothing else changed. Exposure: 3 of 72 synthetic folds.
2. Pre-registered predictions: on the development task above, $G$ wins the cut folds and 3 of 4 outer folds pass; the fourth still fails, because the proposer picks a different extension that its held-out demonstration refutes. So the screen for a new pilot is **predicted to fail**.
3. Kill rule: any new false acceptance or native regression on synthetic or adversarial fixtures, or any change outside the exposure set, rejects P2. Then a new, prospectively frozen, blindly scored pilot; the frozen 120-task pilot has **not** been run.

**Not claimed:** an ARC-AGI-2 solve, a score above 185/1,000, semantic invention, domain generality, or results from the separate VDCG project.

## Questions we expect

**Is this just a refinement loop (CEGIS)?** It includes refinement; the added layer is the typed decision of *what* to change. Its value over matched refinement has not been shown on blind real tasks; that is the decisive comparison.

**Why not raise the cap?** Reaching every cut view would change 73 of 96 synthetic searches, mostly through the last-resort proximity view; forcing a view first was harmful.

**Is the growth grammar a hand-written prior?** Yes, and it is named as one; the proposal is to measure acquisition with that prior fixed and every cost counted.